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MNRU School of Engineering & Technology and Artificial Intelligence

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25CSE104 - Software Crafting & Sculpting Lab

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1	Development of Problem Statement.
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4	Study and usage of any Design Phase CASE tool.
5	Preparation and Development of an ER Diagram for a given problem statement using any Design Phase CASE tool.
6	Preparation and Development of Data Flow Diagrams (DFD) including Context Diagram, Level 0 DFD, and Level 1 DFD for a given system using any Design Phase CASE tool.
7	Performing the Structural Design using any Design Phase CASE tool.
8	Performing the Behavioral Design using any Design Phase CASE tool.
9	Study and usage of any Software Testing Tool for performing Functional and/or Automated Testing on a given application. (Selenium, JUnit, TestNG, JMeter, Postman, etc.)
10	Develop test cases for Unit Testing and Integration Testing.
11	Develop test cases for various White Box and Black Box Testing techniques.
12	Develop a Mini Software Engineering case study and prototype design for a modern application

ONLINE EXAM REGISTRATION SYSTEM

1. Development of Problem Statement:

Organizations and educational institutions often face difficulties in managing and tracking exams assigned to employees or students. Manual exam management methods may lead to delays, communication gaps, poor monitoring, and reduced productivity. The Online Exam Registration System is designed to provide an efficient platform for assigning exams, monitoring progress, generating reports, and maintaining proper communication between managers and employees.

The system helps administrators and managers organize workflows efficiently while enabling employees to update their exam status and deadlines.

Objectives

- Simplify exam assignment process
- Improve communication among users
- Monitor exam progress efficiently
- Generate exam completion reports
- Reduce manual work
- Improve productivity and accountability

Scope of the Project

The system can be used in:

- Educational institutions
- IT companies

- Startups, Team management systems

2. Preparation of Software Requirement Specification (SRS)

Document, Design Documents, and Testing Phase related documents.

Introduction

The Software Requirement Specification document defines the functional and non-functional requirements of the Online Exam Registration System.

I. Functional Requirements:

User Registration and Login: Users should create accounts securely.

- Users should log in using username and password.

Role Management: Admin can manage Administrators and Students.

Exam Creation: Administrators can create exams.

- Exams include title, description, deadline, and priority.

Exam Assignment: Exams can be assigned to employees.

Exam Status Tracking: Students can update exam status.

Notification System: Reminder notifications should be sent.

Report Generation: Administrators can generate reports.

Search and Filter: Users can search exams based on status and priority.

II. Non-Functional Requirements

Security: Password encryption should be implemented.

Reliability: System should function without crashes.

Performance: Dashboard should load within 3 seconds.

Availability: System should be accessible 24/7.

Maintainability: Software should be easy to update.

Scalability: System should support increasing users.

III. User Requirements

Admin Requirements

- Manage users
- Generate reports

Administrator Requirements

- Create and assign exams
- Track progress

Student Requirements

- View assigned exams
- Update status

IV. System Requirements

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4GB minimum
- Internet connection

Software Requirements

- Operating System: Windows/Linux
- Frontend: HTML, CSS, JavaScript

- Backend: Python/PHP/Java
- Database: MySQL

V. Interface Specifications:

User Interface

- Login Page
- Dashboard
- Exam Management Interface
- Reports Interface

Hardware Interface

- Desktop/Laptop
- Mobile Devices

Communication Interface

- HTTP/HTTPS communication

VI. Design Documents

Software Architecture

The system follows a layered architecture.

Layers

1. Presentation Layer
2. Business Logic Layer
3. Data Access Layer
4. Database Layer

MVC Architecture

Model

Handles database operations.

View

Handles user interface.

Controller

Handles user requests and business logic.

VII. Design Quality Attributes

Scalability

Supports increasing users and exams.

Security

Provides authentication and authorization.

Maintainability

Easy to modify and update.

Reliability

Ensures consistent performance.

Usability

Simple and user-friendly interface.

VIII. Testing Phase Documents

Test Plan

Objectives

- Verify software functionality
- Detect errors

- Ensure system quality

Types of Testing

Unit Testing

Tests individual modules.

Integration Testing

Tests interaction between modules.

System Testing

Tests complete integrated system.

Acceptance Testing

Tests user satisfaction.

Test Environment

Component	Specification
OS	Windows/Linux
Database	MySQL
Browser	Chrome/Firefox
Backend	Python/Java/PHP

Sample Test Cases

Test Case ID	Input	Expected Output
TC1	Valid Login	Login Successful
TC2	Invalid Password	Error Message
TC3	Create Exam	Exam Created

TC4	Update Status	Status Updated
TC5	Generate Report	Report Generated

3. Preparation of Software Configuration Management (SCM) and Risk Management related documents.

Introduction

Software Configuration Management manages software changes systematically.

SCM Objectives

- Maintain software versions
- Track modifications
- Control changes
- Ensure software integrity

SCM Activities:

Version Control

Tracks code versions.

Change Management

Controls software modifications.

Build Management

Creates executable versions.

Release Management

Handles software deployment.

SCM Tools:

- Git
- GitHub
- GitLab

Risk Management

Introduction

Risk Management identifies and reduces software development risks.

Selected Process Model

Agile Model

Agile is selected because:

- Flexible requirement handling
- Continuous testing
- Faster feedback
- Easy maintenance

RMMM Plan

Risk	Probability	Impact	Mitigation
Requirement Changes	High	Medium	Agile Planning
Server Failure	Medium	High	Backup Servers
Data Loss	Low	High	Daily Backup
Unauthorized Access	Medium	High	Authentication
Network Failure	Medium	Medium	Auto Save

4. Study and usage of any Design Phase CASE tool.

CASE (Computer Aided Software Engineering) tools assist in software design and development.

Features of StarUML

- UML Diagram Creation
- ER Diagram Support
- Code Generation
- Documentation Support⁵

Installation Steps

1. Download StarUML.
2. Install the software.
3. Create a new project.
4. Select diagram type.

Diagrams Supported

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Activity Diagram
- ER Diagram

Advantages

- Faster Design
- Improved Documentation
- Better Accuracy
- Reduced Development Time

Conclusion

CASE tools improve software design quality and productivity.

5. Preparation and Development of an ER Diagram for a given problem statement using any Design Phase CASE tool.

Introduction

ER Diagram represents entities and relationships in the database.

Entities and Attributes

ADMIN

- Admin_ID (PK)
- Name
- Email

TEAM_MEMBER

- Member_ID (PK)
- Name
- Email
- Designation

TASK

- Exam_ID (PK)
- Exam_Name
- Description
- Deadline
- Priority
- Status

PROGRESS

- Progress_ID (PK)
- Percentage

- Remarks
- Update_Date

NOTIFICATION

- Notification_ID (PK)
- Message
- Date

REPORT

- Report_ID (PK)
- Report_Date
- Completed
- Pending
- Overdue

ER Diagram Representation

Admin ----- Assigns ----- Exam

Team member----- Receives ----- Notification

Admin-----Generates----- Report

6. Preparation and Development of Data Flow Diagrams (DFD) including Context Diagram, Level 0 DFD, and Level 1 DFD for a given system using any Design Phase CASE tool.

Introduction

A Data Flow Diagram (DFD) represents the flow of data within the Online Exam Registration System. It shows how information moves between users, processes, and the database.

DFDs are developed in three levels:

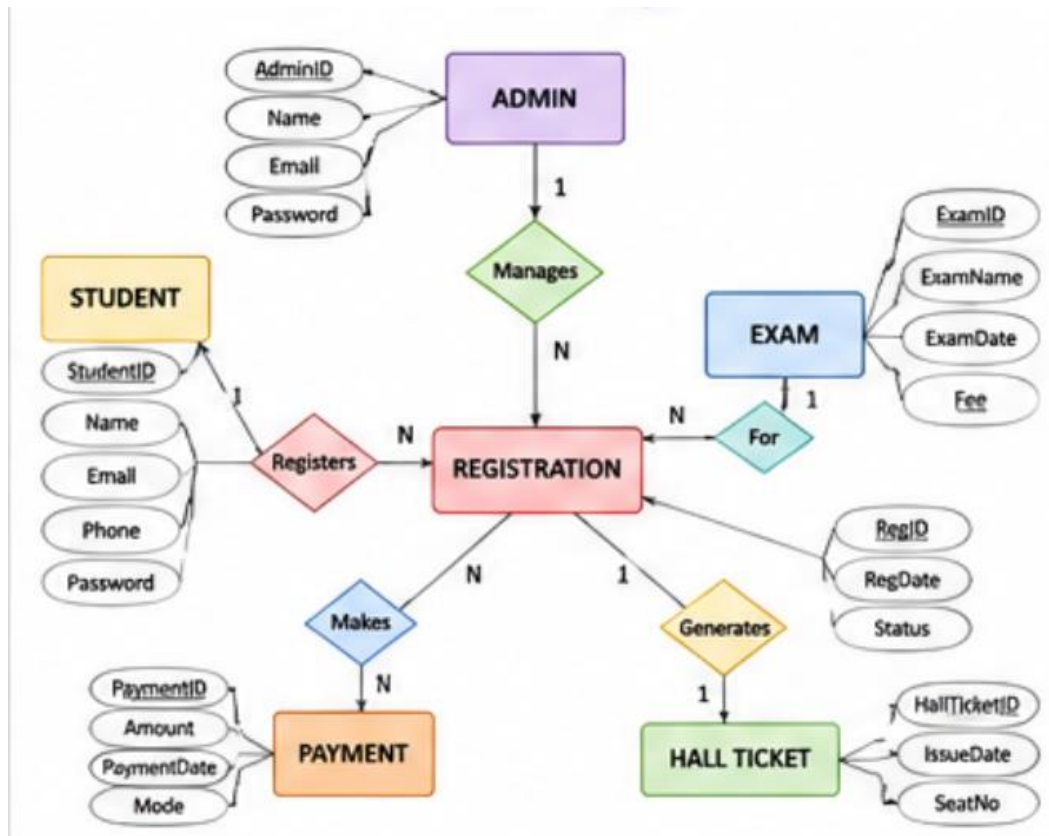
1. Context Diagram
2. Level 0 DFD

3. Level 1 DFD

Context Diagram (Level Context)

Description

The Context Diagram provides a high-level view of the entire system as a single process and shows interactions with external entities.



External Entities

- **Admin**
- Manages users,
- Generates reports



Administrator

- Creates and assigns exams
- Monitors exam progress

Student

- Views assigned exams
- Updates exam status

Data Flow

- Admin sends user management requests.
- Administrator sends exam creation and assignment requests.
- Student sends exam updates.
- Database stores and retrieves all information.

Level 0 DFD

Description

Level 0 DFD breaks the system into major processes.

Main Processes

P1: User Authentication

- Login
- Logout
- User verification

P2: Exam Management

- Create Exam
- Assign Exam
- Modify Exam

P3: Exam Tracking

- Update Status
- Monitor Progress

P4: Report Management

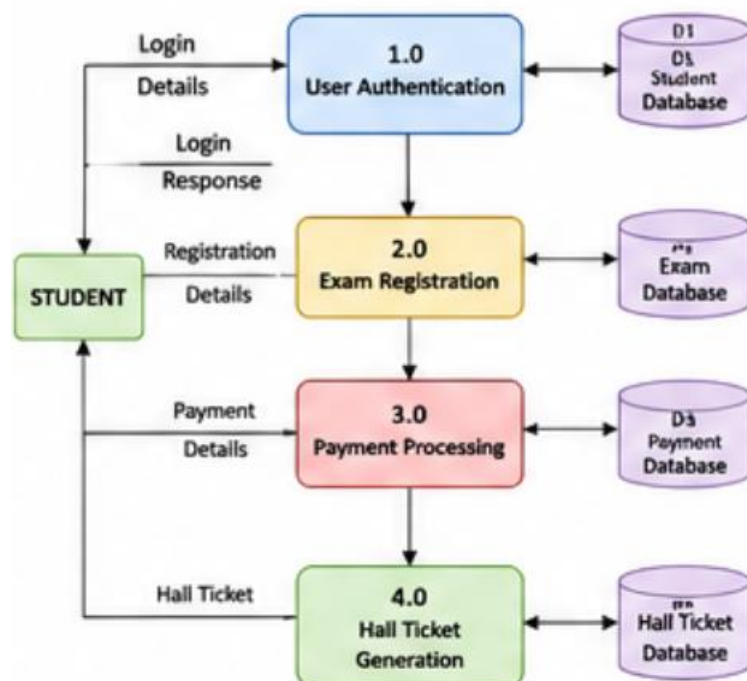
- Generate Reports
- View Reports

Data Stores

D1 – User Database

D2 – Exam Database

D3 – Report Database



Explanation

- User Authentication verifies login credentials.

- Exam Management handles exam creation and assignment.
- Exam Tracking updates progress.

Level 1 DFD

Description

Level 1 DFD further decomposes Process P2 (Exam Management).

Sub Processes

P2.1 Create Exam: Administrator enters exam details.

P2.2 Assign Exam: Exam assigned to employee.

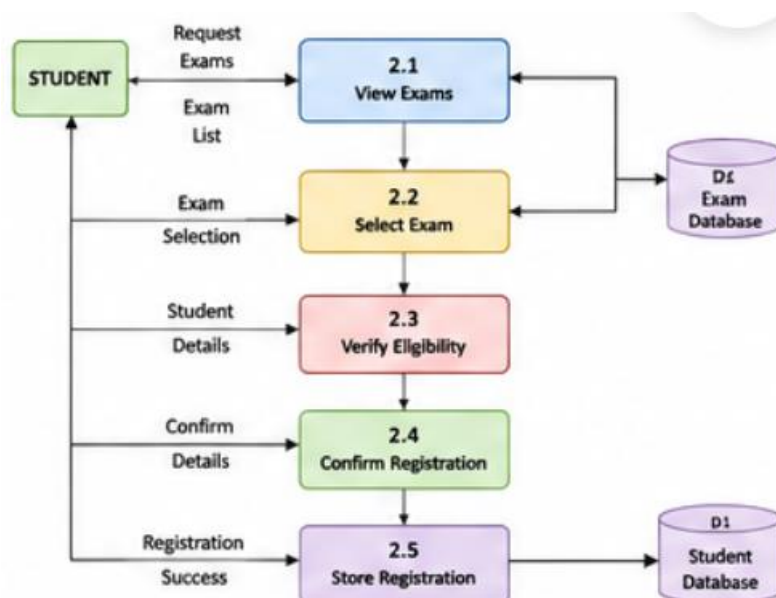
P2.3 Update Exam: Administrator edits exam information.

P2.4 Delete Exam: Administrator removes exam.

P2.5 Store Exam: Exam information stored in datab

Explanation

1. Administrator creates a exam.
2. Exam is assigned to an employee.
3. Exam details can be updated if required.
4. Unnecessary exams can be deleted.
5. All exam information is stored in the Exam Database.



7. Performing the Structural Design using any Design Phase CASE tool.

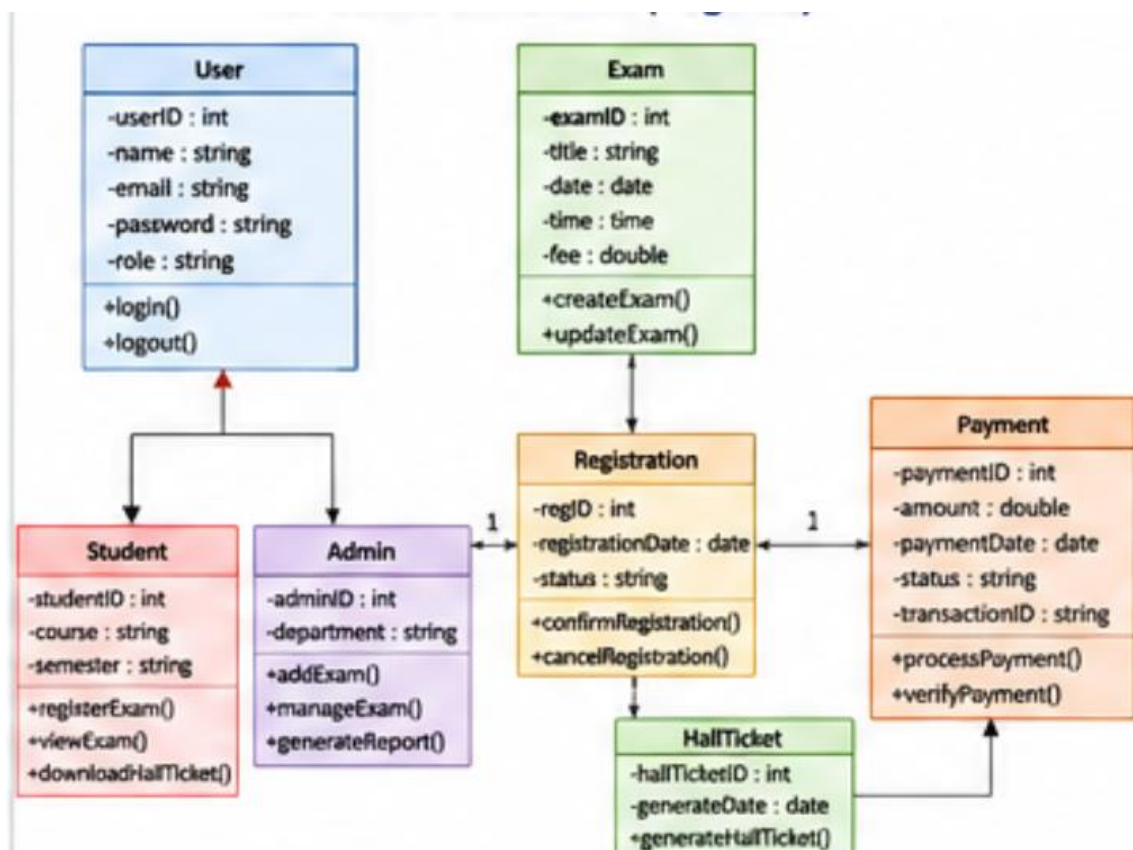
Introduction

Structural design represents static structure of software.

Class Diagram

Classes

- User
- Admin
- Administrator
- Student
- Exam
- Report
- Notification

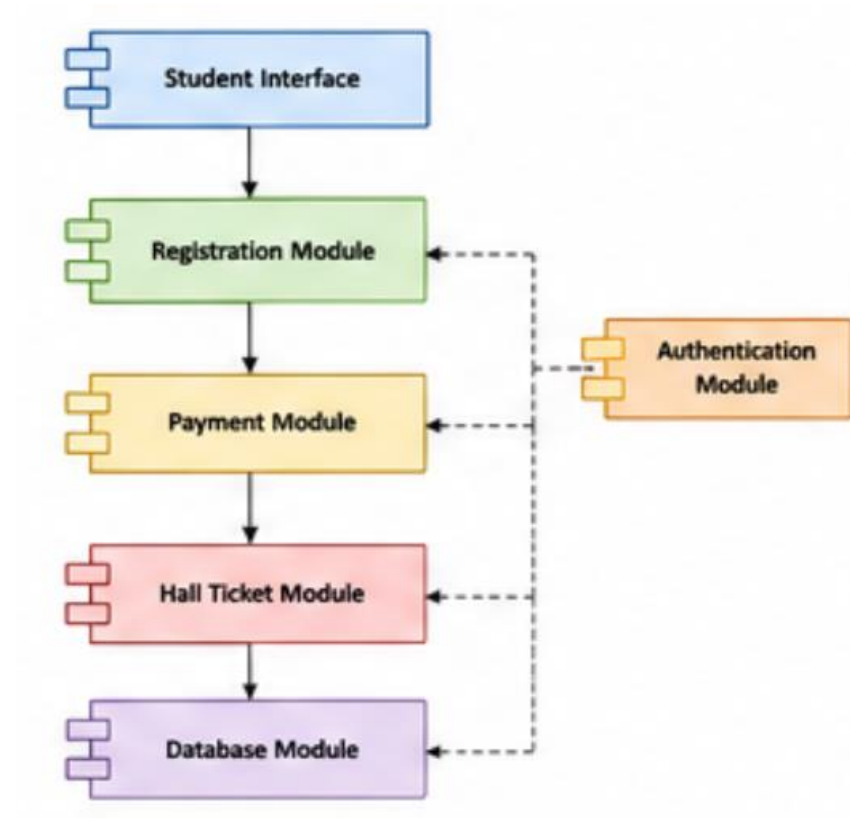


Component Diagram

Purpose

Shows software modules and their dependencies.

Explanation: The UI interacts with business modules, which communicate with the database.



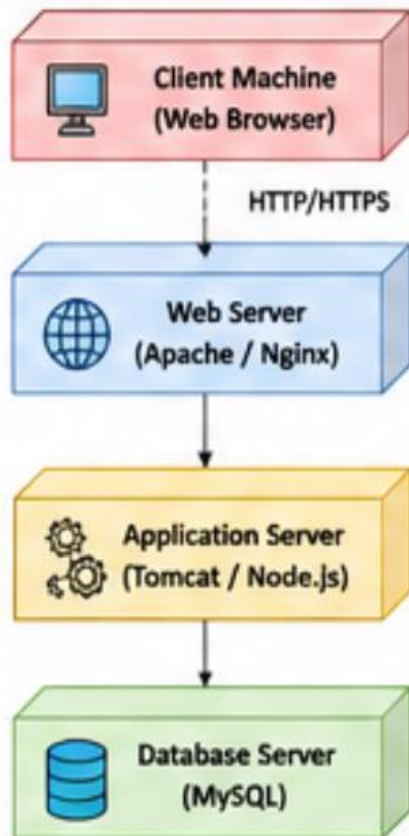
Deployment Diagram

Purpose

Represents physical deployment of hardware and software components.

Explanation

- Users access the system through a browser.
- Application logic runs on the web server.
- Data is stored in the database server.



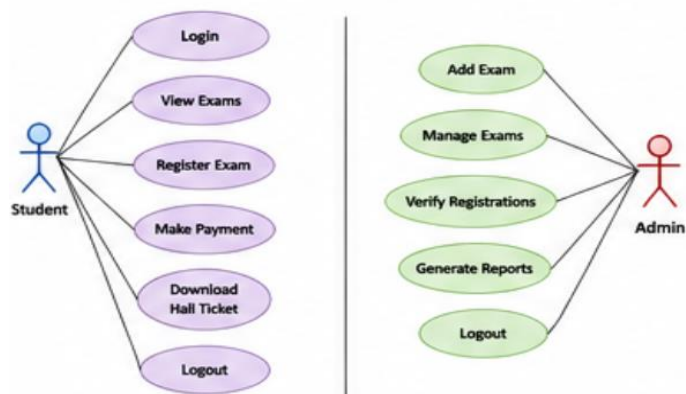
8. Performing the Behavioral Design using any Design Phase CASE tool.

Use Case Diagram

Actors

- Admin
- Student

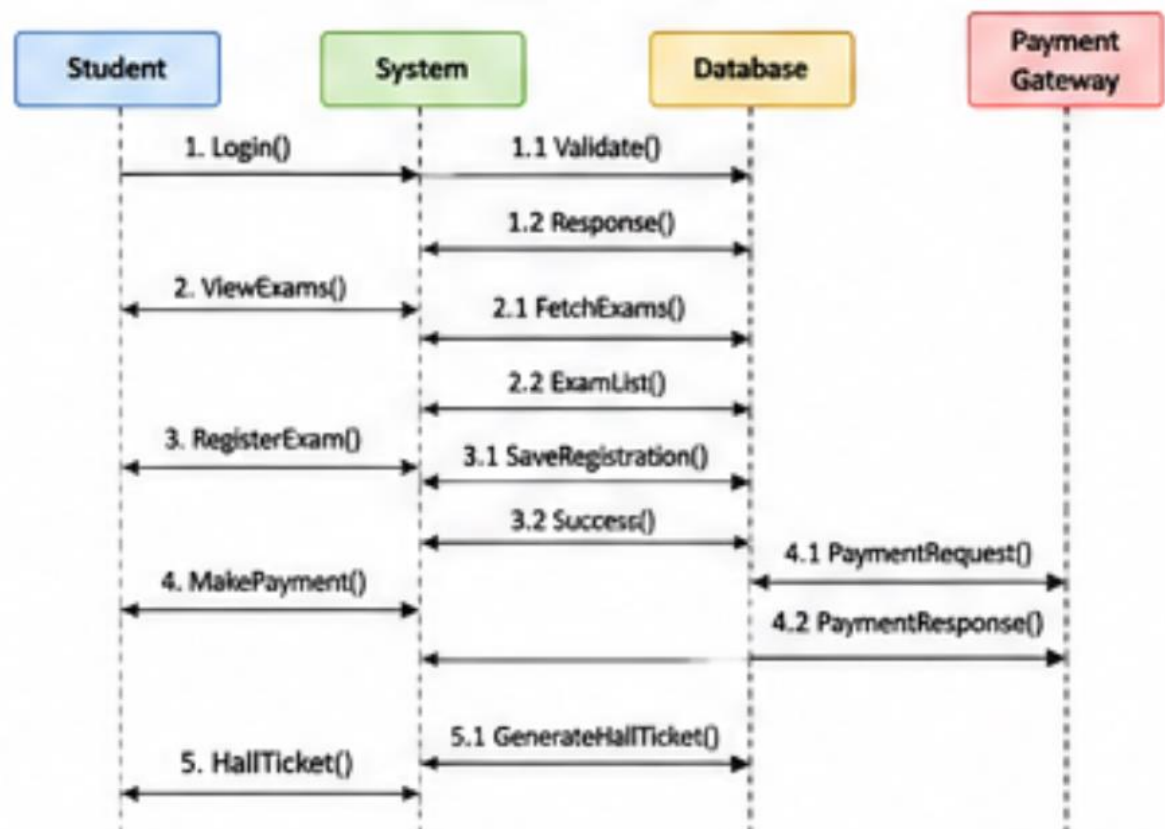
Use Cases: Login, Create Exam, Assign Exam, Update Status, Generate Reports



Sequence Diagram

Objects (Lifelines)

1. Admin
2. System
3. Student
4. Database

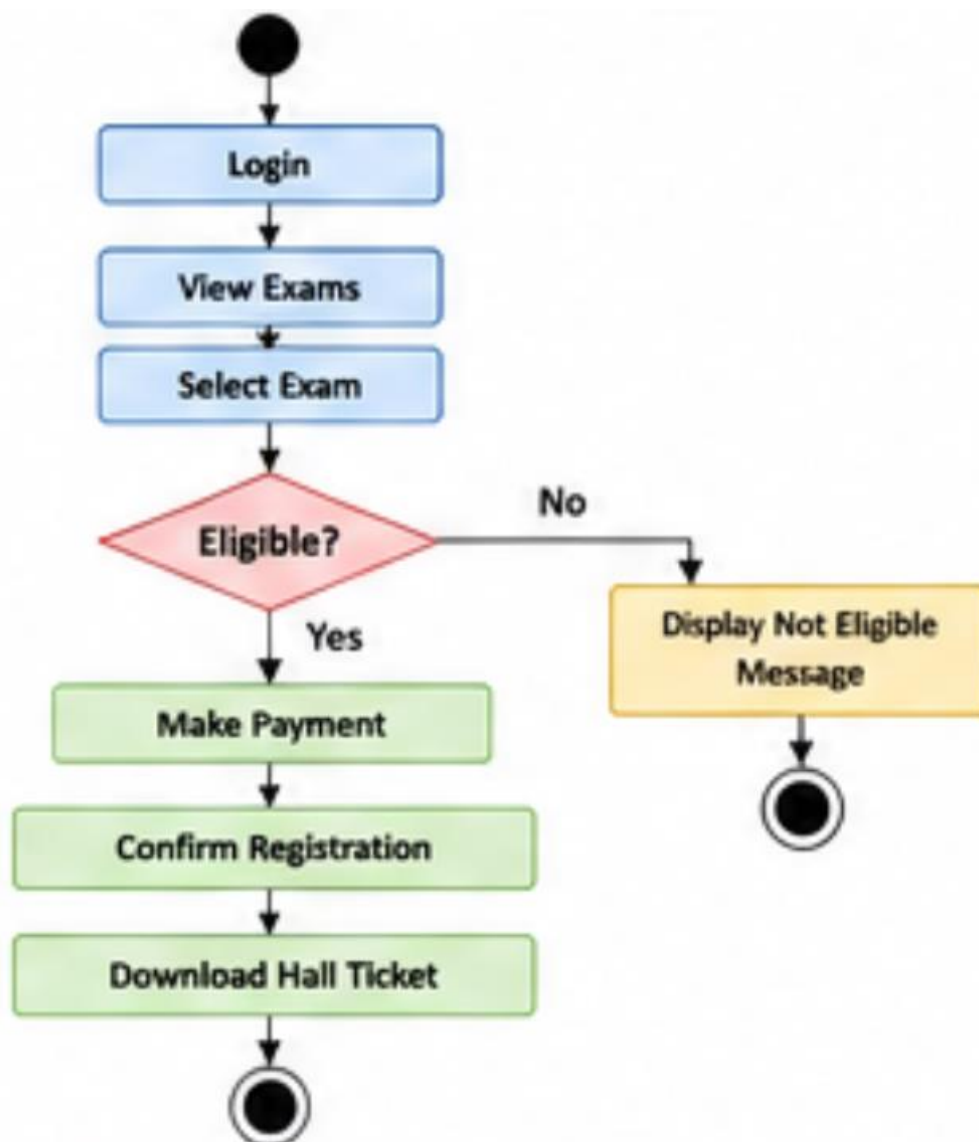


Explanation: Shows the order of interactions when a manager creates and assigns a exam.

Activity diagram

Explanation

Represents the workflow followed by an employee from login to exam completion.

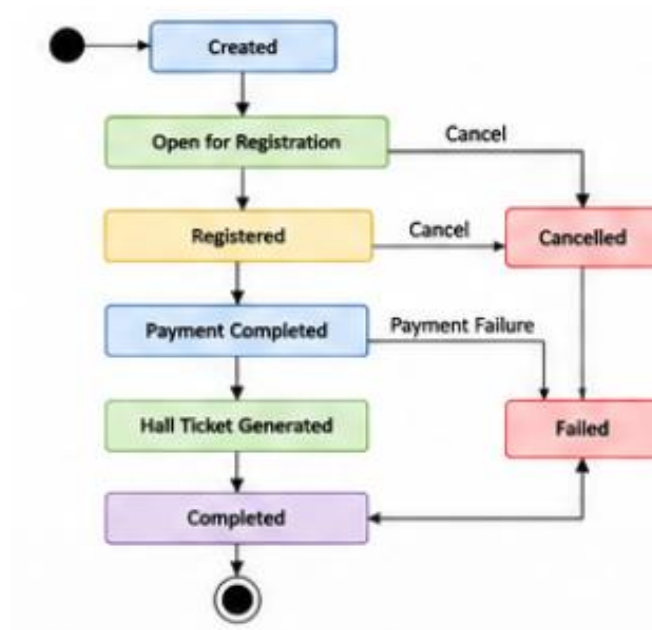


Scenario: Exam Completion Workflow

State chart Diagram

Purpose

Shows different states of a exam throughout its lifecycle.

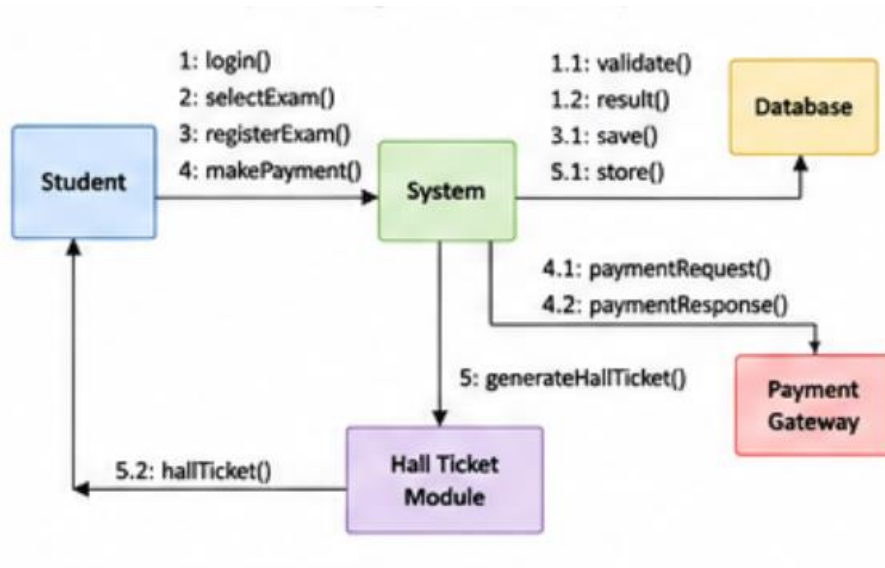


Explanation

Illustrates how a exam changes from one state to another until completion.

Collaboration Diagram

Scenario: Student Updates Exam Status



Explanation

Shows communication among system components and users during exam tracking.

9. Study and usage of any Software Testing Tool for performing Functional and/or Automated Testing on a given application. (Selenium, JUnit, TestNG, JMeter, Postman, etc.)

Testing Tool Used

Selenium

Selenium is used for automated functional testing.

Features of Selenium

- Browser automation
- Functional testing
- Cross-browser support
- Automated test execution

Advantages

- Reduces manual testing effort

- Faster testing
- Reusable test scripts

Example Selenium Testing Activities

- Login Testing
- Form Validation
- Navigation Testing
- Button Click Testing

10. Develop test cases for Unit Testing and Integration Testing.

Unit Testing

Definition

Testing individual modules separately.

Unit Test Cases

Module	Test Scenario	Expected Result
Login	Valid credentials	Success
Exam Creation	Valid exam details	Exam Created
Notification	Reminder trigger	Notification Sent

Integration Testing

Definition

Testing interaction between modules.

Integration Test Cases

Modules Integrated	Expected Result
Login + Dashboard	Dashboard Opens
Exam + Notification	Reminder Sent

Report + Database	Report Generated
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11.White Box and Black Box Testing

Black Box Testing

Definition

Tests functionality without checking internal code.

Black Box Test Cases

Input	Expected Output
Correct Login	Access Granted
Invalid Password	Error Message
Empty Form	Validation Warning

White Box Testing

Definition

Tests internal logic and code paths.

White Box Test Cases

Module	Logic Tested
Login Module	Authentication Logic
Exam Module	Database Storage
Notification Module	Trigger Conditions

ONLINE EXAM REGISTRATION SYSTEM

Requirements Engineering

A.Functional requirements:

Functional requirements describe what the system should do.

User Registration and Login

- Users can create accounts and log in securely.

Role Management

- Different roles such as Admin, Administrator, and Student.

Exam Creation

- Administrators can create exams with title, description, deadline, and priority.

Exam Assignment

- Exams can be assigned to employees or students.

Exam Status Tracking

- Users can update exam status:
 - Pending
 - In Progress
 - Completed

Report Generation

- Generate reports of completed and pending exams.

B.Non-functional requirements:

Non-functional requirements describe how the system should perform.

1. Performance

- System should load dashboards within 2–3 seconds.

2. Security

- User authentication and data encryption must be provided.

3. Reliability

- System should function continuously without crashes.

4. Scalability

- System should support increasing numbers of users and exams.

5. Usability

- Interface should be simple and user-friendly.

6. Availability

- System should be available 24/7.

7. Maintainability

- Easy to update and maintain the software.

Software Requirements Specification (SRS)

Project: Online Exam Registration System

1. User Requirements

User requirements describe the services and features expected by users.

1.1 Admin Requirements

- Admin should be able to create and manage user accounts.
- Admin should assign roles such as Administrator and Student.
- Admin should monitor all exams and project activities.
- Admin should generate reports on exam completion and employee performance.

1.2 Administrator Requirements

- Administrator should create exams with deadlines and priorities.
- Administrator should assign exams to employees.
- Administrator should track exam progress and status.

- Administrator should edit or delete assigned exams.
- Administrator should receive notifications about delayed exams.

1.3 Student Requirements

- Student should log in securely.
- Student should view assigned exams.
- Student should update exam status.
- Student should submit completed exams.
- Student should receive deadline reminders and notifications.

2. System Requirements

System requirements describe the technical and operational requirements of the software.

2.1 Functional Requirements

- The system shall provide secure user authentication.
- The system shall allow exam creation and assignment.
- The system shall maintain exam status tracking.
- The system shall generate exam reports.
- The system shall provide notification and reminder services.
- The system shall support search and filtering of exams.

2.2 Non-Functional Requirements

Performance

- System should respond within 3 seconds.

Security

- Passwords should be encrypted.
- Only authorized users should access data.

Reliability

- System should operate continuously without failures.

Scalability

- System should support increasing users and exams.

Maintainability

- Software should be easy to update and debug.

Availability

- System should be available 24/7.

Usability

- Interface should be user-friendly and easy to navigate.

3. Interface Specifications

Interface specifications define interactions between users and the system.

3.1 User Interface

Login Page

- Username field
- Password field
- Login button

Dashboard

- View exam summary
- Notifications section
- Pending and completed exams display

Exam Management Interface

- Create Exam button
- Edit/Delete options
- Status update section

Reports Interface

- View employee performance

- Generate exam reports

3.2 Hardware Interface

- Desktop/Laptop
- Smartphone/Tablet
- Internet connection

3.3 Software Interface

- Operating System: Windows/Linux
- Database: MySQL
- Frontend: HTML, CSS, JavaScript
- Backend: Python/Java/PHP

3.4 Communication Interface

- Internet-based communication
- Email notifications
- Browser-server communication using HTTP/HTTPS

4. Assumptions and Constraints

Assumptions

- Users have internet access.
- Users possess basic computer knowledge.

Constraints

- System depends on server availability.
- High traffic may reduce performance temporarily.

4.1. System modeling:

A. Use Case Diagram

Explanation

A Use Case Diagram represents interactions between users (actors) and the system.

Actors

1. Admin
2. Administrator
3. Student

Use Cases

- Login
- Create Exam
- Assign Exam
- Update Exam Status
- View Exams
- Generate Reports
- Manage Users
- Logout

Use Case Diagram Explanation

- Admin manages users and generates reports.
- Administrator creates and assigns exams.
- Student updates exam status and submits completed exams.
- All users must log in and log out securely.

B. Class Diagram

Explanation

A Class Diagram shows system classes, attributes, methods, and relationships.

Classes

1. User

Attributes:

- userID
- name
- email
- password

Methods:

- login()
- logout()

2. Admin (inherits User)

Methods:

- manageUsers()
- generateReports()

3. Administrator (inherits User)

Methods:

- createExam()
- assignExam()
- monitorProgress()

4. Student (inherits User)

Methods:

- viewExam()
- updateStatus()

5. Exam

Attributes:

- examID
- title
- description
- deadline

- priority
- status

Methods:

- createExam()
- updateExam()
- deleteExam()

6. Report

Attributes:

- reportID
- reportType

Methods:

- generateReport()

Class Diagram Explanation

- User is the parent class.
- Admin, Administrator, and Student inherit properties from User.
- Administrator interacts with Exam class for assigning exams.
- Reports are generated based on exam information.

C. Sequence Diagram

Explanation

A Sequence Diagram shows interaction flow between objects over time.

Scenario: Student Updates Exam Status

Objects

- Student
- System

Database

Sequence Diagram Explanation

1. Student logs into the system.
2. System verifies credentials using the database.
3. Student views assigned exams.
4. Student updates exam status.
5. Database stores updated information

4.2. Context Model

Explanation

A Context Model shows how the system interacts with external entities such as users and other systems.

External Entities

1. Admin
2. Administrator
3. Student
4. Database Server
5. Notification Service

Context Model Explanation

- Admin manages users and monitors activities.
- Administrator creates and assigns exams.
- Student updates exam progress.
- Database stores exam and user information.
- Notification Service sends reminders and alerts.

4.3. Data Model

Explanation

A Data Model represents how data is stored and related within the system.

Main Entities

1. User

Attributes:

- UserID (Primary Key)
- Name
- Email
- Password
- Role

2. Exam

Attributes:

- ExamID (Primary Key)
- Title
- Description
- Deadline
- Priority
- Status
- AssignedTo (Foreign Key)

3. Report

Attributes:

- ReportID (Primary Key)
- ReportType
- GeneratedDate

4. Notification

Attributes:

- NotificationID (Primary Key)
- Message
- Date

- UserID (Foreign Key)

Data Model Relationships

- One Administrator can assign many Exams.
- One Student can have many Exams.
- One User can receive many Notifications.
- Reports are generated based on Exams.

Data Model Explanation

- User table stores all user details.
- Exam table stores assigned exam information.
- Notification table stores reminder and alert messages.
- Report table stores generated reports.
- Relationships ensure proper exam assignment and tracking.

1. Design engineering:

A. Proposed Software Architecture

The Online Exam Registration System follows a structured architecture where different components handle separate responsibilities.

The system consists of:

1. Presentation Layer

- User Interface
- Login Page
- Dashboard
- Exam Management Screens

2. Business Logic Layer

- Exam Assignment Logic
- Progress Tracking
- Notification Handling
- Report Generation

3. Data Access Layer

- Database Operations
- Data Storage and Retrieval

4. Database Layer

- Stores Users, Exams, Reports, and Notifications

Software Architecture Diagram

B. Architectural Pattern

Selected Pattern: MVC (Model-View-Controller)

The Online Exam Registration System uses the **MVC Architectural Pattern**.

MVC Components

1. Model

Handles data and database operations.

Examples:

- User Data
- Exam Data
- Reports
- Notifications

Responsibilities:

- Store and retrieve data
- Apply business rules

2. View

Represents the user interface.

Examples:

- Login Screen
- Dashboard
- Exam List
- Report Page

Responsibilities:

- Display data to users
- Accept user input

3. Controller

Acts as an intermediary between Model and View.

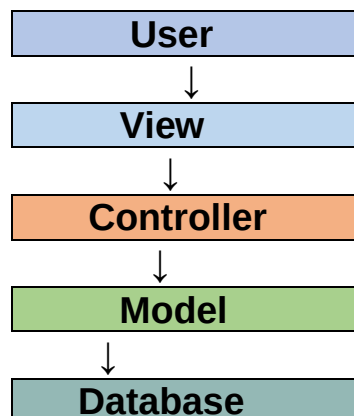
Examples:

- Login Controller
- Exam Controller
- Report Controller

Responsibilities:

- Process user requests
- Update model data
- Return responses to the view

MVC Diagram:



Why MVC is Suitable

- Separates application components clearly.
- Makes development and testing easier.
- Improves maintainability.
- Supports future expansion.

- Simplifies debugging.

C. Design Quality Attributes

1. Scalability

The system can support increasing numbers of users and exams without major performance issues.

Example:

Additional servers or databases can be added when user traffic increases.

2. Maintainability

The system is easy to modify and update.

Example:

Changes in exam management logic can be updated without affecting the user interface.

3. Reliability

The system performs consistently without failures.

Example:

Database backups and error handling improve reliability.

4. Security

Protects user data and system resources.

Features:

- Password encryption
- User authentication
- Role-based access control

5. Performance

The system should respond quickly to user actions.

Example:

Exam lists and dashboards should load within a few seconds.

6. Usability

The system should be simple and user-friendly.

Example:

Easy navigation menus and clear dashboards.

7. Availability

The system should remain accessible most of the time.

Example:

24/7 server availability with backup support.

3. Process and Risk Management:

A. Selected Process Model

Process Model: Agile Model

The Online Exam Registration System uses the Agile Software Development Model.

Why Agile is Selected

Agile is suitable because the project requirements may change during development and continuous improvements are needed.

Advantages of Agile for this System

1. Flexibility

Requirements can be modified easily during development.

Example:

New features like exam reminders or analytics can be added later.

2. Faster Development

The project is developed in small modules called iterations or sprints.

Example:

- Sprint 1 → Login Module
- Sprint 2 → Exam Management
- Sprint 3 → Notifications & Reports

3. Continuous Testing

Testing is performed after every iteration.

Benefit:

Errors are identified early.

4. Better User Feedback

Administrators and employees can give feedback during development.

Benefit:

Improves system usability.

5. Easy Maintenance

Small updates can be added without affecting the entire system.

B. Risk Management**What is Risk Management?**

Risk Management identifies possible problems during software development and prepares solutions to reduce their impact.

RMMM Plan

RMMM stands for:

- Risk Mitigation
- Risk Monitoring
- Risk Management

Risk ID	Risk Description	Probability	Impact	Mitigation Strategy	Monitoring Method	Management Plan
R1	Requirement changes during development	High	Medium	Use Agile methodology	Regular client meetings	Update sprint planning
R2	Server failure	Medium	High	Use backup servers	Server health monitoring	Restore from backup
R3	Unauthorized access or hacking	Medium	High	Implement authentication & encryption	Security audits	Password reset and recovery
R4	Data loss	Low	High	Daily database backup	Backup verification	Restore database backup
R5	Network failure	Medium	Medium	Auto-save functionality	Network monitoring	Retry connection automatically
R6	Delay in project completion	Medium	Medium	Proper sprint planning	Weekly progress review	Allocate extra resources
R7	System performance issues	Low	Medium	Optimize database queries	Performance testing	Upgrade server resources

Risk Explanation

1. Requirement Changes

Users may request additional features after development starts.

Solution:

Agile supports flexible requirement updates.

2. Server Failure

Server crashes may stop system access.

Solution:

Backup servers and cloud hosting.

3. Security Risks

Hackers may attempt unauthorized access.

Solution:

Secure login and encrypted passwords.

4. Data Loss

Database corruption may occur.

Solution:

Regular backups and recovery mechanisms.

5. Network Problems

Internet issues may interrupt exam updates.

Solution:

Auto-save and retry mechanisms.

2. Testing strategy:

A. Test Plan

Introduction

The test plan defines the testing process used to verify that the Online Exam Registration System works correctly and satisfies user requirements.

Objectives of Testing

- Verify all modules work properly.
- Identify and fix software defects.
- Ensure system reliability and security.
- Validate system performance and usability.

Testing Types Used

1. Unit Testing

Tests individual modules separately.

Example:

- Login Module
- Exam Creation Module

2. Integration Testing

Tests interaction between modules.

Example:

Exam Assignment module interacting with Notification module.

3. System Testing

Tests the complete integrated system.

Example:

Complete workflow from login to exam completion.

4. Acceptance Testing

Ensures the software satisfies user requirements.

Example:

Administrator verifies exam assignment functionality.

Testing Environment

Component	Specification
Operating System	Windows/Linux
Database	MySQL
Browser	Chrome/Firefox

Frontend	HTML/CSS/JavaScript
Backend	Python/Java/PHP

B. Black Box Test Cases

Explanation

Black Box Testing checks system functionality without examining internal code.

Black Box Test Cases Table:

Test Case ID	Input	Expected Output	Result
BB1	Valid username & password	Login successful	Pass
BB2	Invalid password	Error message displayed	Pass
BB3	Create exam with all details	Exam created successfully	Pass
BB4	Submit empty exam form	Validation warning displayed	Pass
BB5	Update exam status	Status updated successfully	Pass
BB6	Generate report	Report generated correctly	Pass

C. White Box Test Cases

Explanation

White Box Testing checks internal program logic and code paths.

White Box Test Cases Table

Test Case ID	Module	Logic Tested	Expected Result
WB1	Login Module	Authentication conditions	Correct login validation
WB2	Exam Module	Exam creation flow	Exam stored in database
WB3	Notification Module	Reminder trigger logic	Notification sent
WB4	Report Module	Report generation logic	Accurate report output

WB5	Database Module	Data retrieval queries	Correct records fetched
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D. Validation Approach

What is Validation?

Validation ensures the developed software satisfies user needs and requirements.

Validation Methods

1. Requirement Validation

Checks whether all user requirements are implemented.

Example:

Verify managers can assign exams successfully.

2. Functional Validation

Checks system features and functionality.

Example:

Verify employees can update exam status.

3. User Acceptance Validation

Users test the software in real-world scenarios.

Example:

Administrators test exam tracking workflow.

E. System Testing Approach

What is System Testing?

System Testing verifies the complete integrated system.

System Testing Process

Step 1: Login Testing

Verify secure authentication.

Step 2: Exam Management Testing

Verify exam creation, assignment, editing, and deletion.

Step 3: Notification Testing

Verify reminders and alerts are delivered correctly.

Step 4: Report Testing

Verify reports are generated accurately.

Step 5: Performance Testing

Verify system response speed under multiple users.

Step 6: Security Testing

Verify protection against unauthorized access.

Expected Outcomes

- System functions correctly.
- Data is stored securely.
- Exams are assigned and tracked accurately.
- Reports are generated properly.
- Users can operate the system smoothly.

3. Metrics and Quality:

A. Software Metrics

What are Software Metrics?

Software metrics are measurements used to evaluate the quality, performance, and efficiency of software development and testing.

1. Design Metric

Metric Name: Coupling

Definition:

Coupling measures the dependency between software modules or classes.

Goal:

Lower coupling improves maintainability and flexibility.

Example:

The Exam module should work independently without affecting the Notification module.

Advantage:

- Easier maintenance
- Better modularity
- Simplified debugging

2. Code Metric

Metric Name: Lines of Code (LOC)

Definition:

LOC measures the total number of lines written in the program.

Purpose:

Used to estimate software size and development effort.

Example:

- Login Module → 150 LOC
- Exam Management Module → 300 LOC

Advantage:

- Helps estimate coding complexity
- Measures productivity

3. Testing Metric

Metric Name: Defect Density

Definition:

Defect Density measures the number of defects found per module or per thousand lines of code.

Formula

Defect Density = Amount of Defects / Size of Software

Example:

5 defects found in 1000 lines of code.

Advantage:

- Measures software quality
- Helps improve testing efficiency

Additional Metrics

Code Coverage

Measures percentage of code tested.

Test Case Pass Percentage

Measures percentage of passed test cases.

B. Quality Assurance Activities

What is Quality Assurance (QA)?

Quality Assurance ensures the software meets required standards and functions correctly.

QA Activities

1. Requirement Review

Checks whether requirements are complete and correct.

Example:

Verify all exam management features are included in SRS.

2. Design Review

Checks software architecture and UML diagrams.

Example:

Review MVC architecture design.

3. Code Review

Examines source code for errors and coding standards.

Example:

Check exam assignment logic implementation.

4. Testing Activities

Perform different testing methods.

Examples:

- Unit Testing
- Integration Testing
- System Testing

5. Bug Tracking

Record and monitor software defects until resolved.

Example:

Track login authentication errors.

6. Documentation Review

Verify reports and technical documents are clear and complete.

Benefits of QA Activities

- Improves software reliability
- Reduces defects
- Enhances user satisfaction
- Ensures project quality
-

C. Review Techniques

What are Review Techniques?

Review techniques are methods used to evaluate software documents, designs, and code before testing.

1. Peer Review

Definition:

Team members review each other's work.

Purpose:

Identify mistakes early.

Example:

One developer reviews another developer's exam module code.

2. Walkthrough

Definition:

Author explains the document or code step-by-step to reviewers.

Purpose:

Improve understanding and identify issues.

Example:

Explaining UML diagrams to students.

3. Inspection

Definition:

Formal and detailed review process performed systematically.

Purpose:

Detect defects and ensure quality standards.

Example:

Inspecting test cases and SRS documentation.

4. Technical Review

Definition:

Experts review technical correctness of software components.

Example:

Reviewing database schema design.

Importance of Review Techniques

- Detect defects early
- Improve software quality
- Reduce development cost
- Improve maintainability

4.Conclusion:

The Online Exam Registration System is a user-friendly software solution that streamlines exam management, assignment, and progress tracking for teams and organizations. It enables administrators to assign exams, monitor performance, and enhance communication.

Built using the MVC architecture and the Agile process model, the system ensures flexibility, scalability, and maintainability. Key software engineering concepts applied include requirements engineering, system modeling, design, risk management, testing strategies, and quality assurance. UML diagrams (Use Case, Class, Sequence, Context, Data Models) clearly represent workflows, while risk analysis and RMMM planning strengthen reliability.

Testing methods like Black Box and White Box validate functionality, and software metrics with QA activities improve performance and quality. Overall, it is a secure, scalable application that enhances productivity while demonstrating practical software engineering principles.